

TF-GNN: A temporal feedback graph neural network with discriminative regularization for heterogeneous fraud detection

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HIGHLIGHTS

- Proposes relationship-aware temporal attention for integrating semantic-temporal info in heterogeneous relations.
- Designs a Time Feedback Manager (TFM) that dynamically adjusts neighbor grouping with multi-scale historical predictions.
- Introduces a discriminative regularizer to refine embeddings, boosting intra-group consistency and inter-group separability.

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ABSTRACT

The complexity of Dynamic Graph Fraud Detection (DGFD) stems primarily from the dynamic evolution of fraudulent behavior and graph heterogeneity. Fraudsters typically adjust their strategies over time to evade detection, while spoofed relationships mask obvious fraud patterns. Since existing methods often struggle to accurately capture these risk trajectories and effectively isolate complex heterogeneous interactions, we propose the TF-GNN scheme to address these challenges. TF-GNN incorporates a Temporal Feedback Manager (TFM) utilizing sparsely updated multi-scale LSTM to extract stable temporal features from historical risk sequences. These features generate soft grouping weights to guide dynamic intra-group graph convolutions, followed by adaptive fusion via multi-head attention. Furthermore, a discriminative regularization mechanism optimizes intra-group compactness and inter-group separability. Extensive experiments on real-world financial datasets demonstrate that TF-GNN consistently outperforms strong baselines in AUC and AP metrics, exhibiting robust performance in dynamic environments.

1. Introduction

Graph Neural Networks (GNNs) have emerged as a powerful framework for modeling complex relational structures and have achieved significant success in diverse domains such as electronic payment anti-fraud [1], recommendation systems [2,3], and natural language processing [4,5]. Since the introduction of the Graph Convolutional Network (GCN) [6], a wide range of GNN variants have been proposed, including GraphSAGE [7], Graph Attention Networks (GAT) [8], and others. These models generally follow the unified *message passing* paradigm: each node iteratively aggregates feature information from its neighbors along graph edges and updates its representation through nonlinear transformations. This paradigm is typically grounded in the assumption of *network homophily*—that is, connected nodes tend to share similar features or labels. However, in high-risk domains such as financial fraud detection and cybersecurity, this assumption often breaks down.

For instance, fraudulent nodes may frequently interact with legitimate accounts, leading to heterogeneous connectivity patterns across node types. Such structural heterogeneity severely limits the generalization capability of traditional GNNs, necessitating the development of more robust and heterogeneity-aware architectures.

The complexity of fraud detection tasks not only stems from the heterogeneity of graph structures but is also profoundly reflected in the intricate temporal evolution characteristics of fraud behavior patterns. From a temporal perspective, fraudulent entities, such as fraudulent nodes and malicious merchants, dynamically adjust their strategies based on the degree of risk exposure. In the early stages of fraudulent activities, they deliberately set reasonable transaction amounts and regular login frequencies to reduce the probability of detection. In the later stages, they concentrate large-scale abnormal behaviors, such as batch false transactions or short-term high-frequency interactions, to

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maximize profits. This distinct shift from incubation to outbreak exhibits clear temporal evolution dynamics. Structurally, real-world fraud networks are often heterogeneous graphs containing multiple types of nodes and edges. Different relationship types play complementary roles in capturing fraudulent features, yet fraudulent relationships often mask their abnormal characteristics through complex interactions across multiple relations and views, making these patterns highly concealed.

To address the temporal evolution and structural heterogeneity of fraud, scholars have proposed spectral and spatial methods. Spectral methods focus on the distribution characteristics of graph signals in the frequency domain, enhancing sensitivity to high-frequency abnormal patterns by adjusting the low-to-high frequency component ratio. Methods such as AMNet [9], BWGNN [10], and GHRN [11] design spectral filters for this purpose. Spatial methods perform feature aggregation directly within the node neighborhood. CARE-GNN [12], PC-GNN [13], and RioGNN [14] reduce noise propagation by prioritizing neighbors with similar features through edge trimming or sampling. H2-FDetector [15] explicitly distinguishes homogeneous from heterogeneous edges during aggregation to retain comprehensive features. While these methods have improved the modeling of heterogeneous fraud graphs, several critical deficiencies remain: (1) Non-additivity of node attributes: Direct summation of neighbor features can lead to information loss when features conflict or are distorted. (2) Lack of fine-grained neighbor grouping: Most methods aggregate all neighbors indiscriminately, ignoring the distinct information transmission dynamics between secure and fraudulent nodes. (3) Insufficient temporal utilization: Few methods explicitly integrate temporal sensitivity into multi-relational interactions, failing to capture the combined influence of relational semantics and temporal evolution.

The recent DGA-GNN [16] method alleviates partial false aggregation through a dynamic neighbor grouping strategy and accounts for attribute non-additivity. However, its grouping relies on the prediction results of a single round, making it susceptible to deviations under short-term abnormal fluctuations or prediction noise. Furthermore, DGA-GNN does not introduce a time-sensitive mechanism for relationship types in the temporal modeling of multi-relational interactions, limiting its effectiveness against complex, multi-stage fraudulent behaviors.

To tackle the aforementioned limitations, we propose a novel framework named TF-GNN. We design a relationship-aware temporal attention mechanism to capture heterogeneous interactions. Furthermore, we introduce a Time Feedback Manager (TFM) to dynamically model node risk evolution and guide neighbor grouping. Finally, a discriminative regularization mechanism is employed to enhance the separability of node embeddings. These three designs enhance structural modeling, temporal feedback, and embedding optimization at three distinct levels, effectively alleviating the information pollution problem caused by heterogeneity and significantly improving the model's robustness and interpretability under dynamic and complex fraud patterns. Interpretability is crucial for financial security systems to understand the underlying rationale behind why a specific transaction or user is flagged. In our proposed TF-GNN, the Time Feedback Manager (TFM) naturally provides temporal interpretability by tracking dynamic risk evolution trajectories, while the discriminative regularization explicitly offers representation interpretability by uncovering how camouflaged fraudsters are structurally separated from benign users. The main contributions of this paper are:

1. Proposing a relationship-aware temporal attention mechanism that effectively integrates the semantic information and temporal sensitivity of heterogeneous relations, thereby enhancing cross-relation temporal modeling capability.
2. Designing a Time Feedback Manager (TFM) that dynamically adjusts neighbor grouping by utilizing multi-scale historical prediction information, thereby enhancing the model's adaptability to behavioral evolution.

3. Introducing a discriminative regularization mechanism to enhance intra-group consistency and inter-group separability from the perspective of embedding space optimization.

Experiments conducted on multiple real financial fraud detection datasets demonstrate that TF-GNN is significantly superior to existing state-of-the-art methods in terms of accuracy, robustness, and generalization under dynamic and noisy conditions.

2. Related work

Graph Anomaly Detection (GAD) is designed to identify anomalous entities that exhibit significant deviations from predominant nodal or structural patterns within a graph; consequently, it has garnered extensive attention in applications such as network security, financial risk control, and social public opinion monitoring [17,18]. Based on the target entity, GAD methodologies can be classified into node-level, edge-level, and subgraph-level anomaly detection. Node-level anomaly detection is particularly crucial for financial anti-fraud, focusing on identifying nodes that exhibit anomalous attributes or connection patterns. The majority of extant GAD approaches rely on graph embedding learning, discriminating between legitimate and anomalous nodes by modeling their structural and attribute features. However, the inherent characteristics of fraud—characterized by low homophily, high diversity, sparse distribution, and dynamic temporal evolution—pose significant challenges to the cross-graph generalization and temporal robustness of these methods. In dynamic heterogeneous graph scenarios, the presence of diverse node relationships, complex interaction semantics, and the potential for fraud patterns to be hidden within long-term, multi-relational evolution further diminishes the effectiveness of conventional GAD approaches.

Numerous structural modeling optimization methods have been proposed for fraud detection in practical applications like online finance and social platforms. GraphConsis constructs structural similarity based on consistency to alleviate information inconsistency during label propagation. CARE-GNN [12] introduces a neighbor selector with similar perception to suppress the spread of camouflaged behavior. PC-GNN [13] focuses on neighbor sampling according to label distribution to mitigate data skew. For heterogeneous graphs, H2-FDetector [15] trains an edge classifier to identify connection types and guides the attention mechanism accordingly. GHRN [11] employs pruning strategies to remove heterophilic edges, enhancing semantic expression. BWGNN [10] designs an expressive encoder to improve robustness against structural disturbances under the “right-shift” phenomenon. GAGA [19] integrates group aggregation and learnable coding to adaptively reconstruct the graph structure. CLFS-GNN [20] proposes a two-stage GNN fraud detection model that combines camouflage recognition and a dual-sampling module to handle heterogeneity and imbalance issues. CLFS-GNN utilizes contrastive learning to optimize the feature and structure representation of multi-relationship graphs to enhance the accuracy of fraud detection. Metapath-guided GNN [21] guides heterogeneous graphs to learn financial fraud patterns through metapaths. Although these methods have made progress in camouflage and heterogeneous processing, they ignore multi-scale temporal feedback, resulting in limitations in dynamic evolution scenarios. Other studies tackle the “distribution shift” between training and test data. GDN [22] reveals the impact of dataset statistical differences and proposes a feature separation mechanism for enhanced stability. In fields such as financial fraud (Motie and Raahemi [23]), Sadia Sharmin and Zakia Zaman [24] and fake comments Yu et al. [25], scholars have proposed a variety of structural modeling optimization methods. Consis-GAD [26] mines homogeneous patterns to improve recognition rates while maintaining structural consistency. SEC-GFD [27] uses spectral analysis to aggregate information from multiple frequency bands. PMP [28] differentiates spatial aggregation for homogeneous and heterogeneous neighbors.

Despite these advances, many methods primarily consider feature-level similarity and fail to fully exploit the rich similarity information embedded in the graph structure itself. This insufficiency makes it difficult for models to handle the “heterophily” prevalent in fraud scenarios—where similar nodes may lack connections, while dissimilar nodes form disguised links. This interference undermines GNN aggregation strategies built on the homophily assumption, ultimately reducing classification performance.

The temporal evolution of fraudulent behavior necessitates dynamic modeling. Early dynamic graph methods adopted a time-slicing strategy, partitioning continuously evolving graphs into a series of static snapshots TGAT, and DySA. This approach struggles to capture temporal continuity and is highly sensitive to the chosen window size. To better utilize temporal information, Recurrent Neural Networks (RNNs) and their variants, particularly Long Short-Term Memory (LSTM) networks, have been widely integrated into dynamic graph modeling. EvolveGCN [29] dynamically updates GCN weights via LSTM to adapt to structural changes. TGN [30] combines message passing with time encoding, using GRU/LSTM to maintain node states for event-level modeling. Building upon these sequence-modeling foundations, recent advancements in graph-based fraud detection have increasingly emphasized the joint modeling of spatial and temporal dimensions. Highly sophisticated architectures such as Spatial-Temporal Gated Networks (STGN) [31] and Spatial-Temporal-Aware Graph Transformers (STA-GT) [32] have been proposed. These state-of-the-art methods excel at learning complex sequential patterns and transaction motifs by leveraging advanced spatial-temporal gating and transformer mechanisms.

Despite these advancements, existing dynamic and spatial-temporal methods still exhibit key limitations when confronted with deep camouflage in broad heterogeneous networks. First, their temporal modeling is often single-scale and lacks a historical prediction feedback mechanism, making it difficult to simultaneously capture short-term sudden behaviors and long-term evolutionary trends. Second, while effective for sequential transactions, they often fail to actively disentangle highly camouflaged heterophilic nodes, where fraudsters deliberately establish connections with benign entities to obscure their structural signatures. To address these distinct challenges, our proposed TF-GNN diverges

from standard spatial-temporal aggregation. It integrates a Multi-Scale Temporal Feedback Manager (TFM) to overcome the limitations of single-scale LSTMs, coupled with a Discriminative Regularization mechanism to actively separate camouflaged nodes in complex heterogeneous environments.

Furthermore, as the field rapidly evolves, the advancement of Large Language Models (LLMs) has opened new orthogonal avenues for fraud detection, particularly on text-rich graphs. Recent works have explored integrating LLMs with graph structures to leverage their powerful semantic understanding capabilities. For instance, FLAG [33] proposes an LLM-enhanced GNN framework that utilizes LLMs to extract discriminative text to mitigate neighborhood camouflage, while DGP [34] a graph-enhanced LLM approach employs dual-granularity prompting to process text-rich and numerical graph neighborhoods. While our proposed TF-GNN primarily focuses on resolving the challenges of temporal trajectory evolution and structural heterophily, these recent LLM-based approaches highlight the critical value of deep semantic mining. Integrating LLM-driven semantic extraction with TF-GNN’s dynamic structural modeling presents a highly promising direction for our future research.

3. Method

To address the temporal drift of node behavior, the evolution of relational structures, and the uncertainty associated with noisy neighbors, the TF-GNN architecture is proposed herein. This architecture incorporates relation-aware temporal attention, TFM, dynamic group aggregation, multi-head attention fusion, and discriminative regularization, as illustrated in Fig. 1.

3.1. Neural network architecture

Given a dynamic heterogeneous graph $\mathcal{G}$ and the node feature matrix $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ (where d is the feature dimension), as well as the historical behavior sequence $\mathcal{H} = \{H_v^{(t)} | v \in \mathcal{V}, t \in \mathcal{T}\}$, the task objective is to learn a mapping function to predict the label of each node:

$$f : (\mathcal{G}, \mathbf{X}, H_v^{(t-1)}) \rightarrow p_v^{(t)}. \quad (1)$$

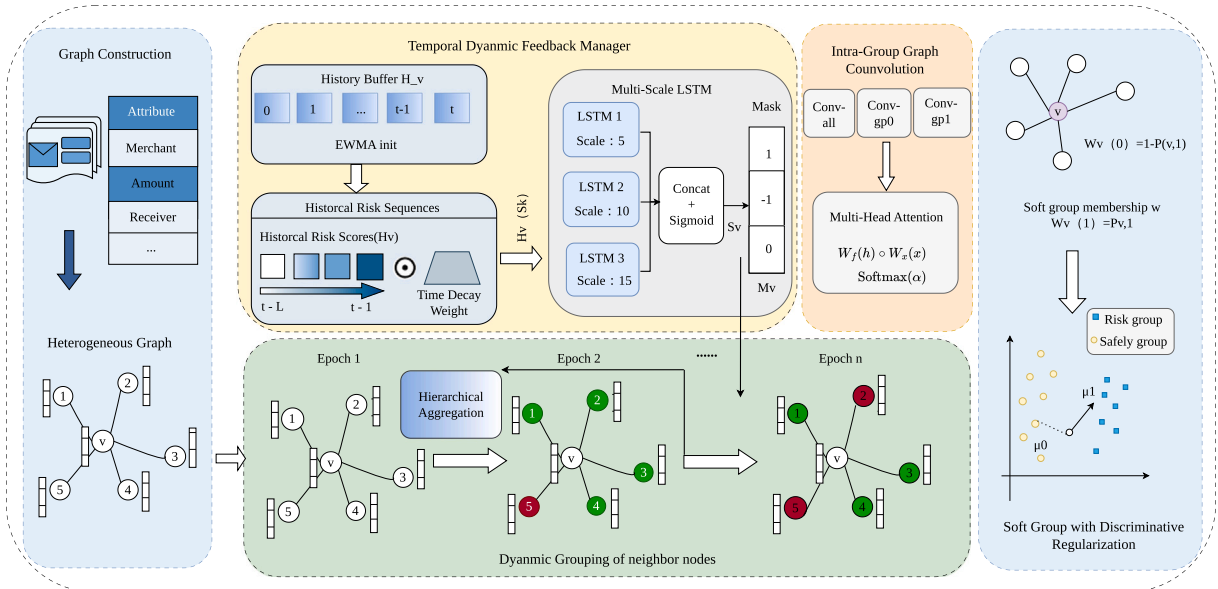


Fig. 1. The TF-GNN model consists of six core modules. *Graph Construction* builds a heterogeneous transaction graph that includes multiple types of nodes and edges such as accounts, merchants, and transactions; the *History Buffer + EWMA* maintains and smooths the historical risk scores of nodes; *Multi-Scale LSTM* captures short-term, medium-term and long-term risk evolution patterns; *Mask* generates group masks based on risk scores; *Dynamic grouping + group Conv* dynamically divides nodes according to the mask and performs differentiated graph convolution and multi-head attention aggregation within the group; *Soft group Regularization* enhances representational discriminability by minimizing intra-group variance and maximizing inter-group distance.

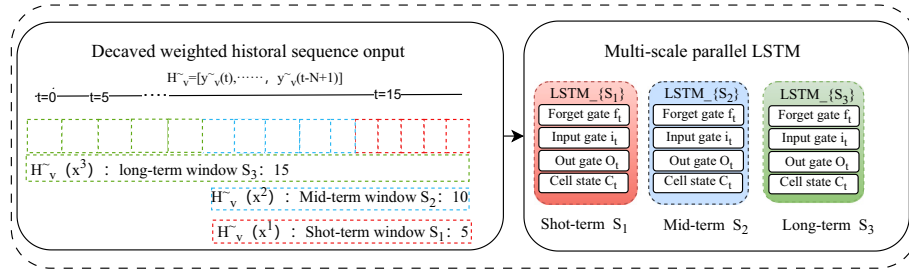


Fig. 2. Multi-Scale LSTM temporal pattern extraction.

Here, $p_v^{(t)} = P(y_v^{(t)} = 1 | \mathcal{G}, \mathbf{X}, H_v^{(t-1)}) \in [0, 1]$ is the probability that node v is a fraudulent entity at time t . The historical prediction sequence for node v is denoted as $H_v^{(t-1)} = [p_v^{(t-L)}, p_v^{(t-L+1)}, \dots, p_v^{(t-1)}]$. $p_v^{(t)} = 1$ indicates that node v exhibits fraudulent behavior at time t , while $p_v^{(t)} = 0$ indicates it is a normal entity.

3.2. Time feedback manager (TFM)

To overcome the limitations of traditional GNNs regarding static neighbor grouping, the Temporal Feedback Manager (TFM) is introduced. Different from the two-stage camouflage recognition GNN proposed by Guan et al., the TFM utilizes historical prediction trajectories to delineate the risk evolution of nodes, subsequently guiding neighbor grouping to achieve temporal adaptability. Specifically, the TFM integrates EWMA with multi-scale coding to output a node-level risk score $s_v \in [0, 1]$ and construct a soft grouping prior mask $M_v = [1 - s_v^{(t)}, s_v^{(t)}] \in \mathbb{R}^2$. To enhance both robustness and efficiency, a sparse update strategy is employed.

For a given node's historical prediction sequence $H_v \in \mathbb{R}^L$, where $H_v = [h_{v,1}, h_{v,2}, \dots, h_{v,L}]$ and $h_{v,i}$ represents the predicted probability of node v at the i -th historical moment, the following strategies are applied: **Exponential Weighted Moving Average (EWMA)**: Controls the smoothness of historical predictions.

$$h_{v,i} = \alpha \cdot p_v^{(0)} + (1 - \alpha) \cdot h_{v,i-1}, \quad i = 1, 2, 3, \dots, L. \quad (2)$$

Here, $p_v^{(0)}$, the initial predictor of node v , is obtained through a simple MLP. $\alpha \in (0, 1)$ is the EWMA attenuation coefficient, controlling the degree of influence of the initial value. It is selected via cross-validation to balance short-term fluctuations and long-term trends.

Sparse update mechanism: The historical sequence is updated only when the prediction probability change is significant, to suppress instability caused by noise and reduce computational burden.

$$H_v^{(t)} = \begin{cases} \text{CircularShift}(H_v^{(t-1)} \circ [p_v^{(t)}]) & \text{if } |p_v^{(t)} - p_v^{(t-1)}| > \tau, \\ H_v^{(t-1)} & \text{otherwise.} \end{cases} \quad (3)$$

$$\text{UpdateFreq}_v = \frac{1}{T} \sum_{t=1}^T \mathbb{I}[|p_v^{(t)} - p_v^{(t-1)}| > \tau]. \quad (4)$$

CircularShift($\cdot$) denotes the circular left shift operation (deleting the oldest element), and $\circ$ is the vector concatenation operation. τ is the update threshold, which defines an update frequency index to measure the instability of node behavior. This mechanism effectively filters out small fluctuations and improves the stability of cache updates.

Time Decay Weighting: To emphasize recent behavioral patterns, exponential time decay weights are assigned to the historical sequence by the TFM.

$$\tilde{H}_v = H_v \odot \omega, \quad (5)$$

$$\omega_i = \frac{\exp(-\lambda(L - i + 1))}{Z}, \quad i = 1, 2, \dots, L. \quad (6)$$

Here, $\odot$ represents the Hadamard product (element-wise multiplication), Z is a normalization constant, and λ is the attenuation rate. The

resulting weight distribution is monotonically increasing ($\omega_1 < \omega_2 < \dots < \omega_L$) assigning the maximal weight to the most recent timestamp, thereby enhancing the model's responsiveness to behavioral shifts.

Multi-scale LSTM coding: To augment the model's capacity for capturing both short-term fluctuations and long-term trends, a multi-scale LSTM architecture is employed to perform multi-granularity modeling of the weighted historical sequence. Fig. 2 Let $S = \{S_1, S_2, \dots, S_K\}$ be a set of time window lengths representing multiple time scales. Each scale S_k corresponds to an independent LSTM network. The weighted sequence $\tilde{H}_v$ is processed to generate subsequences for each scale, which are then fed into their respective LSTMs:

$$(o_k, h_k, c_k) = \text{LSTM}_{S_k}(\tilde{H}_v^{(S_k)}), \quad (7)$$

where $o_k \in \mathbb{R}^{S_k \times d_h}$ is the output sequence, and $h_k, c_k \in \mathbb{R}^{d_h}$ are the final hidden and cell states, respectively. The hidden states from all scales are concatenated and passed through a fully connected layer to obtain the final time risk score:

$$f_v = \text{concat}([h_1, h_2, h_3]) \in \mathbb{R}^{3d_h}, \quad (8)$$

$$s_v^{(t)} = \sigma(\mathbf{W} \cdot f_v + \mathbf{b}) \in [0, 1]. \quad (9)$$

This mechanism allows the model to capture fraud signals emerging at different temporal ranges.

3.3. Multi-channel attention fusion and feature aggregation

For each node v , three distinct feature representations are constructed by aggregating information from neighbors in different semantic groups: **All Neighbors** ($\mathbf{h}_v^{\text{all}}$): Aggregates information from all types of neighbors to capture the global structural context. **Safe Group** ($\mathbf{h}_v^{g0}$): Aggregates information from neighbors predicted to belong to the low-risk (safe) group. **Risk Group** ($\mathbf{h}_v^{g1}$): Aggregates information from neighbors predicted to belong to the high-risk group.

An intra-group convolution (IntraConv) is performed on these three channels, resulting in a matrix representation:

$$\mathbf{H}_v = \begin{bmatrix} \mathbf{h}_v^{\text{all}} \\ \mathbf{h}_v^{g0} \\ \mathbf{h}_v^{g1} \end{bmatrix} \in \mathbb{R}^{3 \times d}. \quad (10)$$

A lightweight multi-head attention mechanism is then adopted for adaptive fusion. For each attention head h :

$$\mathbf{Q}_h = \mathbf{H}_v \mathbf{W}_Q^{(h)}, \quad \mathbf{K}_h = \mathbf{H}_v \mathbf{W}_K^{(h)}, \quad (11)$$

$$\mathbf{A}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{3 \times 3}, \quad (12)$$

$$\mathbf{Z}_h = \mathbf{A}_h \mathbf{H}_v \in \mathbb{R}^{3 \times d}. \quad (13)$$

The outputs of all heads are concatenated and transformed to generate the final fused node representation. To prevent over-smoothing

and encourage sparsity, L1 regularization on the attention weights is introduced:

$$\mathcal{L}_{\text{attention}} = \lambda_{\text{sparse}} \sum_{v,h} \|\alpha_{v,h}\|_1, \quad (14)$$

where λ_{sparse} is an adjustable hyperparameter. The learned attention weights provide interpretability regarding the model's reliance on different group representations.

3.4. Dynamic grouping aggregation and fusion

To synthesize current predictions with historical time-series modeling, a weighted fusion mechanism is utilized to compute a comprehensive score determining the final grouping:

$$s_v^{\text{final}} = \beta \cdot p_v^{(t)} + (1 - \beta) \cdot s_v^{(t)}. \quad (15)$$

Here, β serves as a fusion weight, balancing the relative significance of current versus historical information. A three-way grouping is then applied:

$$\text{Grouping Label}(v) = \begin{cases} 1 & \text{if } s_v^{\text{final}} > \theta_2, \\ -1 & \text{if } \theta_1 < s_v^{\text{final}} \leq \theta_2, \\ 0 & \text{if } s_v^{\text{final}} \leq \theta_1. \end{cases} \quad (16)$$

3.5. Soft group loss with discriminative regularization

To enhance the discriminability of node representations, a Soft Grouping mechanism is proposed, introducing two discriminative regularization terms to optimize intra-group compactness and inter-group separability, as shown in Fig. 3. First, the centers of the two soft groups are calculated:

$$\mu_k = \frac{\sum_{v=1}^N w_v^{(k)} \mathbf{h}_v}{\sum_{v=1}^N w_v^{(k)} + \epsilon}, \quad k \in \{0, 1\}, \quad (17)$$

where $w_v^{(k)}$ is the soft weight for node v in group k , and $\mathbf{h}_v$ is its embedding. The regularization terms are defined as follows:

Intra-group consistency loss:

$$\mathcal{L}_{\text{intra}} = \mathcal{L}_0 + \mathcal{L}_1 = \sum_{k \in \{0,1\}} \frac{\sum_{v=1}^N w_v^{(k)} \|\mathbf{h}_v - \mu_k\|^2}{\sum_{v=1}^N w_v^{(k)} + \epsilon}. \quad (18)$$

Inter-group separation loss (using cosine distance):

$$\mathcal{L}_{\text{inter}} = 1 - \frac{\mu_0^\top \mu_1}{\|\mu_0\|_2 \cdot \|\mu_1\|_2 + \epsilon}. \quad (19)$$

The total soft grouping loss is a weighted sum of these two terms:

$$\mathcal{L}_{\text{soft}} = \beta \mathcal{L}_{\text{intra}} + \omega \mathcal{L}_{\text{inter}}, \quad (20)$$

where β and ω are equilibrium coefficients.

3.6. Dynamic scheduling and two-stage training strategy

To avoid interference from unstable early predictions, a dynamic scheduling function is used to gradually increase the weight of the soft grouping loss:

$$\gamma(t) = \begin{cases} 0 & t < T_{\text{warmup}} \\ \gamma_{\text{max}} \cdot \sigma\left(5 \frac{t - T_{\text{warmup}}}{T_{\text{max}} - T_{\text{warmup}}} - 2.5\right) & t \geq T_{\text{warmup}} \end{cases}, \quad (21)$$

where $\sigma(\cdot)$ is the sigmoid function, T_{warmup} is the warm-up period, and γ_{max} is the maximum loss weight.

A two-stage training strategy is adopted to enhance model stability:

1. **Phase One:** Freeze GNN parameters and optimize only the TFM. The objective is to stabilize its time modeling and learn an effective risk scoring mechanism. The loss function is the standard cross-entropy loss based on the TFM output.

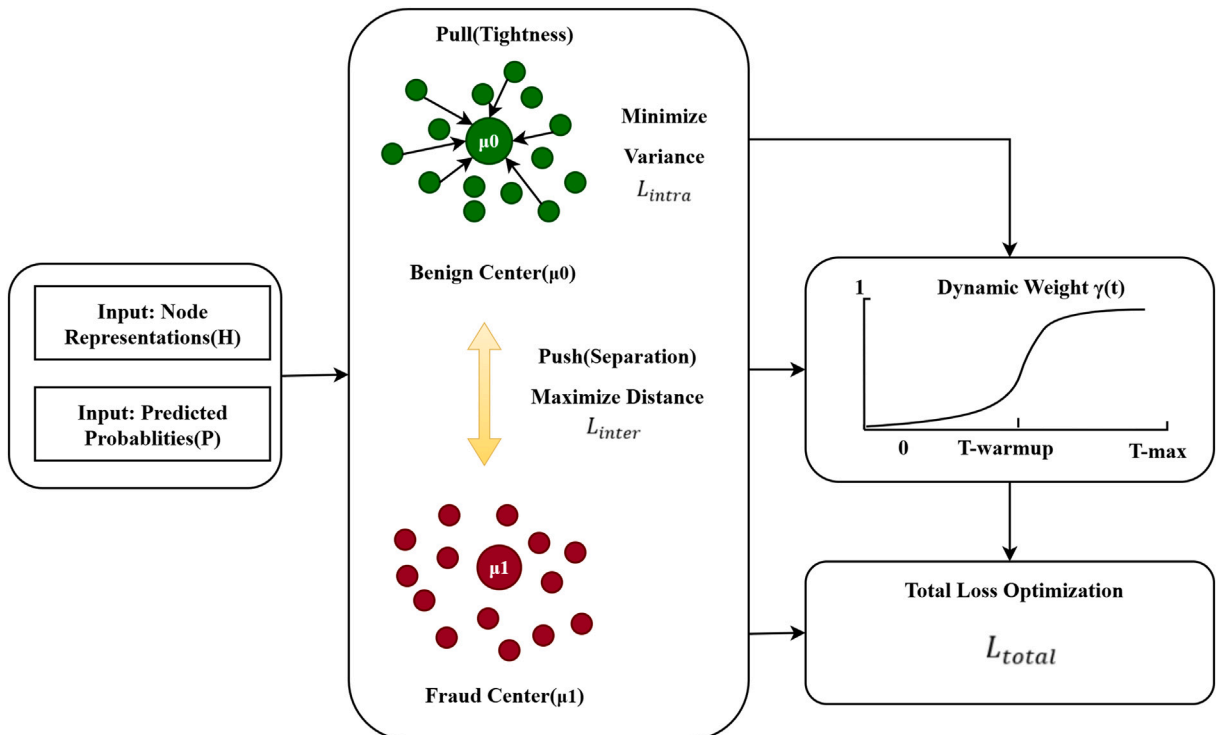


Fig. 3. Framework of the soft group loss with discriminative regularization for enhanced embedding separability.

2. **Phase Two:** Unfreeze all modules and perform end-to-end fine-tuning. The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \gamma(t)\mathcal{L}_{\text{soft}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \quad (22)$$

where $\mathcal{L}_{\text{ce}}$ is the cross-entropy loss, $\mathcal{L}_{\text{soft}}$ is the dynamically scheduled soft grouping loss, and $\mathcal{L}_{\text{reg}}$ is an L2 regularization term with weight λ_{reg} .

3.7. Complexity analysis

Although TF-GNN incorporates multiple modules, its computational complexity remains linear and highly scalable. The Temporal Feedback Manager (TFM) processes individual node sequences with an $\mathcal{O}(N \times L)$ complexity (N nodes, sequence length L), which is further minimized by the sparse update mechanism (Eq. 4) that skips redundant calculations. For structural aggregation, the multi-channel convolution requires $\mathcal{O}(E \times d)$ operations (E edges, hidden dimension d), while the channel-level attention avoids the typical $\mathcal{O}(N^2)$ bottleneck. Finally, Discriminative Regularization only incurs training overhead, adding zero cost during inference. Overall, TF-GNN's total inference complexity is strictly bounded by $\mathcal{O}(N \times L + E \times d)$, ensuring substantial performance gains without over-parameterization.

Algorithm 1 TF-GNN training procedure.

Require: Dynamic heterogeneous graph $G = (V, E, R, T)$

- 1: Node features $X \in \mathbb{R}^{|V| \times d}$
- 2: Labels $y \in \{0, 1\}^{|V|}$
- 3: Hyperparameters: history length L , sparse threshold τ , EWMA factor α ,
- 4: fusion weight λ , regularization schedule $\gamma(t)$, freeze epochs T_{freeze}

Ensure: Trained model parameters Θ

▷ Initialization

- 5: Initialize all network parameters Θ
- 6: Initialize history buffers H_v (length L) for all nodes
- 7: Initialize EWMA states s_v^{ewma}
 - ▷ Stage 1: TFM Stabilization (freeze GNN, train TFM only)
- 8: **for** epoch = 1 **to** T_{freeze} **do**
- 9: **for** each batch B **do**
- 10: Forward frozen GNN $\rightarrow$ temporary predictions (for history only)
- 11: EWMA + sparse update H_v and s_v^{ewma} ($|p_v^{(t)} - s_v^{\text{ewma}}| > \tau$)
- 12: Multi-scale LSTM $\rightarrow s_v \leftarrow \text{MultiScaleLSTM}(\tilde{H}_v)$
- 13: Temporary classification loss on s_v
- 14: Update **only** TFM parameters
- 15: **end for**
- 16: **end for**
 - ▷ Stage 2: Full End-to-End Training
- 17: **for** epoch = $T_{\text{freeze}} + 1$ **to** T_{max} **do**
- 18: **for** each batch B **do**
- 19: Full forward pass $\rightarrow q_v$ (current prediction)
- 20: Fuse temporal risk: $\tilde{s}_v = \lambda q_v + (1 - \lambda)s_v$
- 21: Generate super mask: $M_v = [1 - \tilde{s}_v, \tilde{s}_v]$
- 22: Three-channel intra-group convolution $\rightarrow \{h^{\text{all}}, h^{\text{safe}}, h^{\text{risk}}\}$
- 23: Multi-head channel attention fusion $\rightarrow z_v$
- 24: Final prediction: $p_v^{(t)} \leftarrow \sigma(\text{MLP}(z_v))$
- 25: Loss: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \gamma(t)\mathcal{L}_{\text{soft}} + \mathcal{L}_{\text{reg}}$
- 26: Full backpropagation (update all parameters)
- 27: Sparse update history buffer with $p_v^{(t)}$
- 28: **end for**
- 29: Update regularization schedule $\gamma(t)$
- 30: Early stopping on validation AUC
- 31: **end for**
- 32: **return** Trained parameters Θ

Table 1

Statistics of the four fraud detection datasets. *Fraud (%)* indicates the proportion of fraudulent nodes. *#nodes* and *#edges* denote the total number of nodes and edges, respectively. *#features* represents the dimensionality of node attributes.

Dataset	#nodes	#edges	#features	Fraud (%)
T-Finance	39,357	21,222,543	10	4.58
Elliptic	46,564	73,248	93	9.76
YelpChi	45,954	3846,979	32	14.53
Amazon	11,944	4398,392	25	6.87

4. Experiment

4.1. Experimental environment and dataset

To validate the effectiveness of TF-GNN in heterogeneous graph fraud detection tasks, extensive evaluations were conducted on multiple real-world datasets. All experiments were implemented using the PyTorch 2.0 framework on a hardware platform equipped with an NVIDIA RTX 3070 GPU. Each experiment was repeated five times with different random seeds, and the reported results are the average values to ensure statistical reliability. All hyperparameters were meticulously optimized on the validation set through grid search.

Dataset

We selected four representative, real-world financial fraud detection datasets that cover diverse application scenarios, including e-commerce reviews, financial transaction networks, and Bitcoin network analysis. The statistics for these datasets are summarized in Table 1.

Elliptic [35]: This dataset is designed for detecting illegal Bitcoin transactions. It consists of multiple temporal subgraphs, which pose challenges for model training due to the presence of isolated nodes in the training and validation sets, potentially affecting static grouping strategies.

T-Finance: A dataset for detecting financial transaction fraud. Nodes represent accounts and transactions, while edges represent transfer relationships between them.

YelpChi [36]: A dataset for detecting false reviews, belonging to a multi-relational graph. The dataset contains review information for hotels and restaurants on Yelp, where vertices are reviews and edges represent relationships between them. Spam comments constitute the primary anomaly.

Amazon [37]: A dataset for detecting false product reviews, and also a multi-relationship graph. It represents Amazon's user review network within the musical instruments category, with vertices as users and edges as user-item interaction relationships.

Dataset Preprocessing: For all datasets except Elliptic, we partitioned the data into training, validation, and testing sets using a 4:2:4 ratio. This partitioning was performed using the utility from the scikit-learn package, maintaining a consistent random seed as per previous work. For the Elliptic dataset, partitioning respected the timestamps of transaction entities, aligning with the official recommendations for dataset partitioning.

Evaluation Metrics We employed the `cal_metrics` function from our `myutilspy` utility library to evaluate model performance. Given the highly imbalanced nature of fraud detection tasks, we selected AUC (Area Under the ROC Curve) and AP (Average Precision) as the primary evaluation metrics to comprehensively assess the model's discriminative power and overall detection performance in unbalanced classification scenarios. In addition, we also report the recall of the fraud class, which directly reflects the model's ability to correctly identify minority transactions under severe class imbalance.

4.2. Comparison of baseline model and overall performance

To evaluate the effectiveness of the proposed TF-GNN model in dynamic graph fraud detection, we selected four representative categories

of baseline methods for comparison. These baselines span from traditional machine learning models to state-of-the-art graph neural networks (GNNs) and dedicated methods designed for graph-structured fraud detection tasks, allowing us to assess TF-GNN's advantages from multiple dimensions.

- **Traditional Machine Learning Methods:** We selected two classic, non-graph-based classifiers as performance baselines: Multi-Layer Perceptron (MLP) [38] and Random Forest (RF) [39]. These models rely solely on node attribute features and provide a lower-bound performance reference for scenarios where graph structure information is ignored.
- **General Graph Neural Network Methods:** To investigate the impact of incorporating graph structure, we introduced two standard GNN architectures: Graph Convolutional Network (GCN) [6] and Graph Attention Network (GAT) [8]. GCN performs node representation updates by aggregating information from neighbors using Laplacian smoothing principles. GAT, on the other hand, introduces a self-attention mechanism to assign adaptive weights to neighboring nodes, highlighting the contributions of more important neighbors.
- **Specialized GNNs for Fraud Detection:** To benchmark against advanced, domain-specific methods, we selected two representative models designed for static heterogeneous fraud graphs: PC-GNN [13] and GHRN [11]. PC-GNN employs a location-aware mechanism to screen adjacency relationships, while GHRN combines heterogeneous information modeling with a global graph representation mechanism.
- **Hybrid Models Combining GNN with Traditional ML:** Additionally, we incorporated two hybrid fraud detection models that combine traditional machine learning with GNN architectures: DGA-GNN [16] and GAGA [19]. DGA-GNN adapts to dynamically changing graph structures through a group-based structure-aware mechanism. PMP [40], which addresses graph heterophily by applying distinct message passing strategies to separate label-consistent and label-inconsistent topologies.

To ensure a fair comparison, this paper obtained the source code for the comparison methods PC-GNN, GHRN, DGA-GNN, GAGA and PMP from the official documentation and ran these models using the authors' recommended default parameter settings. For Multilayer Perceptron (MLP) (Hinton 1990) and Random Forest (RF) (Breiman 2001), we implemented MLP and RF using the Scikit-learn toolbox. Traditional graph neural network algorithms, namely Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT), were implemented using the DGL framework to eliminate performance differences caused by data splitting. All experiments were conducted on the same hardware platform to ensure a fair comparison under identical experimental settings. This includes the exact same dataset partitioning, identical random

seeds, and the same hardware platform. This multi-faceted comparison allows us to evaluate TF-GNN's advantages in dynamic graph modeling, structural understanding, and temporal representation learning from multiple dimensions.

4.2.1. Performance comparison

The performance of TF-GNN versus the baseline models was evaluated on the four datasets, with results presented in Table 2. TF-GNN consistently outperformed all baselines across all metrics and datasets, demonstrating significant improvements.

As shown in Table 2, TF-GNN consistently achieves the highest performance across all four datasets and both evaluation metrics (AP and AUC), outperforming all baseline methods. Notably, TF-GNN surpasses the previously state-of-the-art hybrid model, DGA-GNN [16], on these datasets. On the Elliptic dataset, TF-GNN achieves the best AP score (83.37%) with an AUC only slightly higher than DGA-GNN (95.84% vs. 94.24%), demonstrating its exceptional capability in handling the complex temporal and structural challenges of Bitcoin transaction graphs. This comprehensive superiority highlights the effectiveness of TF-GNN's core innovations: the relationship-aware temporal attention, the Time Feedback Manager, and the discriminative regularization mechanism, in capturing dynamic, heterogeneous fraud patterns.

The consistent superiority of TF-GNN across diverse datasets can be attributed to its targeted solutions to the shortcomings of specific baseline models. First, compared to standard static fraud detection models such as PC-GNN that primarily focus on structural features while completely ignoring temporal evolution, TF-GNN maintains a dynamic, multi-scale history buffer through its TFM module. This enables TF-GNN to capture sudden fraudulent behavior that static models cannot detect. Second, compared to recently proposed heterogeneity-aware GNNs such as GHRN and the newly added PMP, TF-GNN exhibits a significant advantage in dynamic environments. While PMP successfully mitigates structural heterogeneity through partitioned message passing, it fundamentally lacks the ability to track dynamic behavior over time. TF-GNN overcomes this deficiency by explicitly combining discriminative regularization with dynamic temporal updates, performing particularly well on datasets with rich temporal interactions. Finally, compared to existing dynamic graph fraud detectors such as DGA-GNN, which lack time-sensitive mechanisms for different relation types, TF-GNN's dynamic grouping mechanism can proactively and adaptively filter out deceptive spoofed edges based on historical risk feedback. This powerful synergy between multi-scale temporal modeling and structural regularization explains why TF-GNN can achieve good results in various complex graph environments. Particularly in terms of the newly introduced Recall metric, TF-GNN demonstrates substantial superiority over all baselines. Given the extreme class imbalance, standard GNNs tend to misclassify disguised fraudsters as benign, leading to high false-negative rates. By employing discriminative regularization to

Table 2

Comparison results (%) of eight methods on four benchmark datasets. AP denotes average precision, AUC denotes the area under the ROC curve, and recall refers to the recall of the fraud class. In each group, the leading score is marked with an underline. The bold values indicate the results achieved by our proposed method.

Method	Metrics	Elliptic			T-finance			YelpChi			Amazon		
		AP	AUC	Recall	AP	AUC	Recall	AP	AUC	Recall	AP	AUC	Recall
Traditional Methods	MLP	61.68	88.12	27.42	66.27	91.61	52.13	46.57	82.35	56.35	63.94	90.50	<u>81.51</u>
	RF	<u>63.37</u>	83.89	<u>59.70</u>	<u>80.66</u>	<u>93.80</u>	<u>59.36</u>	<u>74.30</u>	<u>92.35</u>	40.01	<u>83.74</u>	<u>95.56</u>	80.60
General GNNs	GCN	<u>34.76</u>	80.49	47.23	76.78	91.20	<u>66.48</u>	25.68	<u>62.07</u>	50.81	29.79	80.76	<u>67.49</u>
	GAT	26.75	79.57	<u>55.76</u>	<u>77.82</u>	<u>91.35</u>	50.28	23.60	59.64	<u>54.67</u>	<u>86.16</u>	<u>96.11</u>	65.44
Specialized GNNs	PC-GNN	37.98	<u>87.66</u>	46.84	63.79	87.16	49.41	47.76	81.79	<u>65.74</u>	85.44	96.65	61.42
	GHRN	51.54	83.91	<u>49.73</u>	83.84	<u>95.93</u>	69.38	<u>62.43</u>	86.64	56.76	87.10	<u>97.68</u>	<u>78.72</u>
Combined Methods	GAGA	28.01	82.68	47.68	84.03	96.51	75.23	79.51	93.88	76.35	88.35	96.58	<u>89.27</u>
	DGA-GNN	<u>78.95</u>	<u>94.24</u>	<u>72.29</u>	90.55	97.69	<u>83.51</u>	<u>92.82</u>	<u>97.89</u>	<u>85.10</u>	<u>92.17</u>	<u>98.29</u>	81.21
	PMP	51.89	80.04	50.08	<u>90.74</u>	<u>97.74</u>	79.86	78.14	94.11	77.63	90.48	97.79	84.42
Ours	TF-GNN	83.37	95.84	74.32	91.10	97.59	86.35	94.49	98.23	87.41	93.44	98.62	82.42

explicitly push apart heterophilic connections, TF-GNN successfully uncovers these hidden minority nodes, resulting in a significantly higher Recall.

4.3. Model interpretability and visualization analysis

In the initial stages of training, model predictions exhibit significant fluctuations due to incomplete convergence. As training progresses and the model learns from increasingly stable representations, the frequency of category estimation inversions gradually decreases. This stabilization of predictions leads to a marked improvement in classification accuracy, as illustrated in Fig. 4.

Temporal Interpretability (TFM and Multi-scale LSTM): The TFM, empowered by the multi-scale LSTM, explicitly tracks the dynamic evolution of node behaviors across different time windows. Instead of acting as a black-box aggregator, TFM provides transparent intermediate outputs in the form of continuous risk trajectories (as visualized in Fig. 5). The short-term and long-term scales capture both sudden bursty fraudulent activities and gradual camouflage building. Furthermore, the temporal risk scores generated by TFM directly guide the dynamic grouping masks. This provides structural transparency, explaining exactly *when* and *why* the model dynamically adjusts its neighborhood aggregation strategy in response to evolving fraud patterns.

Representation Interpretability (Discriminative Regularization): On the other hand, the discriminative regularization provides clear spatial interpretability by illustrating *how* the model disentangles camouflaged fraudsters from benign users. The explicit “push-and-pull” mechanism (intra-group tightness and inter-group separation) has a clear mathematical intuition. To validate this, we utilize t-SNE to visualize the node embeddings in Fig. 6. Compared to the original mixed feature space (Figs.6a and 6c), the embeddings

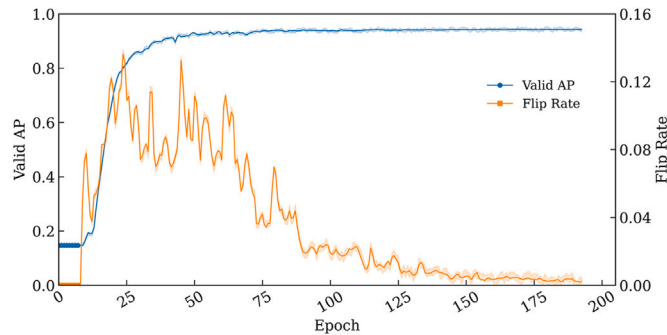


Fig. 4. As the number of training epochs increases, the flip rate of y in the YelpChi dataset and the accuracy of the validation set change.

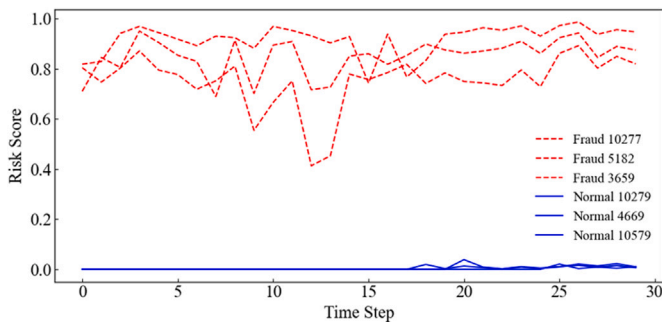


Fig. 5. Temporal Interpretability: time-series risk trajectory of typical nodes. The X-axis represents the chronological time steps, and the Y-axis denotes the predicted fraud risk score bounded between 0 and 1. The explicit peaks in the curves intuitively demonstrate TF-GNN’s ability to capture sudden bursty fraudulent behaviors and trace latent evolutionary camouflage over time.

after applying our regularization (Figs.6b and 6d) show distinctly separated clusters. This visual evidence intuitively explains that the model successfully strips away the structural camouflage, making the decision boundary highly interpretable.

Nodes of different categories form clearly separated distribution areas in the two-dimensional space. Most fraudulent nodes are clustered in specific high-risk clusters, indicating that the model can effectively capture abnormal behavior patterns and achieve discriminative representation of nodes of different categories. Some mixed distribution samples can still be observed in the boundary area. These nodes usually correspond to users with ambiguous behavioral characteristics or in a state of transition. These nodes show abnormal behavior in recent interactions, but have not yet formed significant fraudulent features, resulting in their embedding positions being between normal and fraudulent clusters. By comparing the embedding distributions under different models or different regularization strategies, the contributions of the grouping mechanism and the regularization term in improving the separability of embeddings can be intuitively evaluated.

The temporal risk trajectory and the t-SNE embedding visualization experiment jointly verified the effectiveness of the TF-GNN model in temporal perception and representation learning: The model can not only achieve early warning of fraudulent behavior through the time feedback mechanism, but also form a highly discriminative feature distribution in the embedding space, providing an intuitive explanatory basis for subsequent dynamic grouping optimization and anomaly detection.

4.4. Ablation study

To rigorously evaluate the contribution of each core module in TF-GNN, we conducted a systematic ablation study on the T-Finance dataset. We evaluated the model’s performance after disabling or modifying key components, which yielded the following findings.

Time Feedback Manager (TFM): Removing the TFM results in the most significant performance degradation (-0.99% AUC). The core value of TFM lies in its ability to construct a vertical risk trajectory for each node by leveraging historical prediction sequences. This temporal context is crucial for identifying complex, long-term evolving fraudulent behaviors that models relying solely on instantaneous features would miss. The absence of TFM also leads to a higher group turnover rate, confirming its effectiveness in stabilizing the dynamic neighbor grouping process.

Multi-scale LSTM: To validate the necessity of the multi-scale temporal modeling, we conducted an experiment on the YelpChi dataset by removing the multi-scale design of the LSTM, which resulted in a noticeable drop in AUC. User behavior on this platform exhibits strong long-term periodicity, requiring a synergistic combination of short-term and long-term temporal modeling. As shown in Figs. 7, increasing the number of temporal scales improves performance, albeit with diminishing marginal returns.

Discriminative Regularization: This module aims to enhance the separability of node representations between different risk groups. Removing this regularization term leads to a slight but consistent performance degradation across all datasets. Furthermore, we found that carefully tuning the regularization strength or using a dynamic scheduling strategy can further improve accuracy, demonstrating the effectiveness of applying moderate discriminative constraints during training (Table 3).

4.5. Parameter sensitivity analysis

To verify the robustness and stability of the TF-GNN model with respect to its key hyperparameters, we conducted parameter sensitivity experiments on four datasets. To analyze the impact of the dynamic grouping threshold z on model performance, we calculated the changes in AP and AUC values.

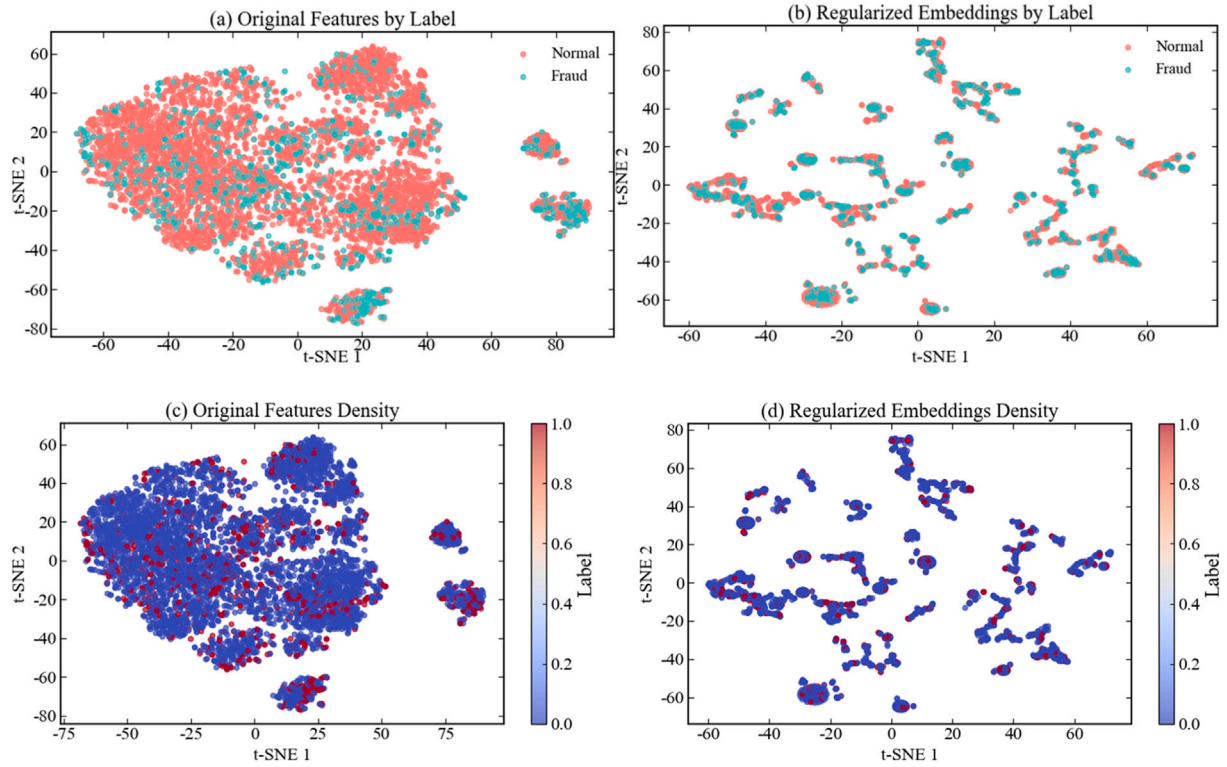


Fig. 6. Representation Interpretability: t-SNE visualization of node embeddings before and after applying discriminative regularization. The red dots represent fraud nodes, and the green/blue dots represent benign nodes. (a) and (c) show the original mixed feature space where fraudsters camouflage themselves among normal users. (b) and (d) demonstrate that our proposed explicit “push-and-pull” mechanism successfully disentangles the structural camouflage, pushing fraud clusters away from benign ones to create a highly interpretable decision boundary.

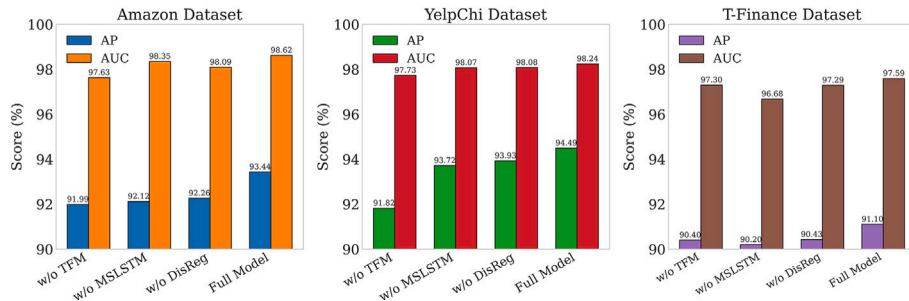


Fig. 7. Ablation study results on the Amazon, YelpChi, and T-Finance datasets. We report Average Precision (AP) and Area under the ROC Curve (AUC) under different model variants (w/o TFM, w/o MS-LSTM, w/o DkReg, and Full Model). The results demonstrate the contribution of each module to fraud detection performance.

Table 3

Ablation study of TF-GNN on Amazon, YelpChi, and T-finance datasets. “w/o TFM”, “w/o MSLSTM”, and “w/o DisReg” denote the removal of the temporal feedback manager, multi-scale LSTM, and discriminative regularization, respectively.

Method	Amazon		YelpChi		T-Finance	
	AP	AUC	AP	AUC	AP	AUC
w/o TFM	91.99	97.63	91.82	97.73	90.40	97.30
w/o MSLSTM	92.12	98.35	93.72	98.07	90.20	96.68
w/o DisReg	92.26	98.09	93.93	98.08	90.43	97.29
TF-GNN (Full)	93.44	98.62	94.49	98.24	91.10	97.59

The threshold z determines the activation intensity of the Time Feedback Manager (TFM) for neighborhood dynamic adjustment. Its role is to control the minimum difference in risk changes among nodes within a group, thereby balancing the temporal sensitivity and grouping

stability of the model. Through the analysis of sensitive parameters, it can be seen that the overall stability is high. The TF-GNN model demonstrates good robustness on all four datasets, as shown in Fig. 8. The figure shows the changes in AP and AUC with the threshold z , which are generally stable without significant fluctuations, indicating that the

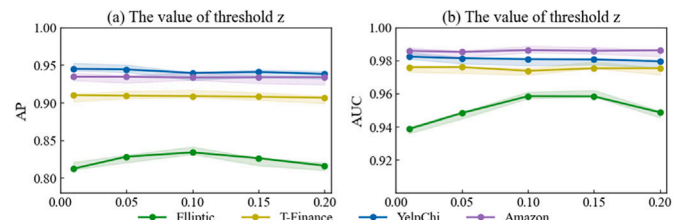


Fig. 8. Parameter analysis of the threshold.

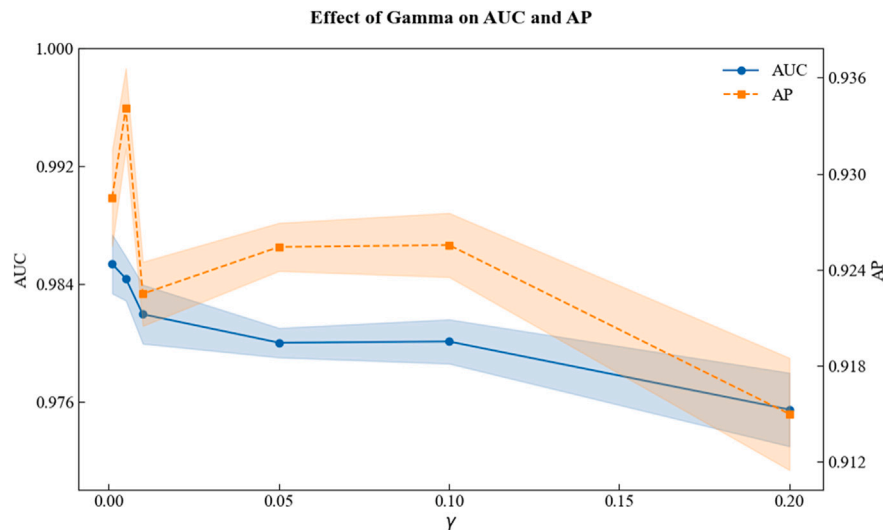


Fig. 9. Regularization coefficient.

model can maintain stable performance under different time grouping granularities.

YelpChi and Amazon performed the best, with their AP and AUC reaching the highest levels across all thresholds. This indicates that in multi-relationship review networks and e-commerce scenarios, the model can fully capture heterogeneous interactions and temporal evolution features. The performance of T-Finance remained generally stable during the change of threshold z from 0.00 to 0.20, without a significant peak. This indicates that the model has low sensitivity to threshold changes in financial trading scenarios and exhibits strong adaptability. Elliptic's performance slightly declines when the threshold increases, especially when $z > 0.10$, where the AUC shows a slight drop. This indicates that in sparse dynamic networks like Bitcoin transactions, an overly large packet threshold may weaken the ability of the time feedback mechanism to capture local structural changes. However, overall, it still maintains a relatively high performance level, reflecting the fundamental stability of the model in heterogeneous sparse networks.

To evaluate the influence of the discriminative regularization coefficient γ on the performance of the TF-GNN model, a sensitivity analysis was conducted on the Amazon dataset. The analysis focused on examining the impact of varying γ within the range $[0, 0.2]$ on the AUC and AP metrics. This parameter controls the relative weights assigned to intra-group compactness and inter-group separability during training. If γ is too small, it may lead to insufficient regularization, resulting in underfitting. Conversely, if γ is too large, it may introduce excessive smoothing or amplify noise, leading to overfitting. The experiment employed a grid search method, with each configuration repeated five times. The reported results are the mean values $\pm$ standard deviation to ensure statistical reliability. As illustrated in Fig. 9, the X-axis represents the γ value, the left Y-axis represents the AUC score, and the right Y-axis represents the AP score. The results reveal that both the AUC and AP curves exhibit a clear trend of rising initially and then declining: At $\gamma = 0.001$, AUC reaches its peak at 0.9862 and AP peaks at 0.9344. At $\gamma = 0.005$, the performance remains at a relatively high level. When $\gamma > 0.01$, both indicators gradually decline. At $\gamma = 0.2$, AUC drops to 0.976 and AP drops to 0.915. This demonstrates that a moderate γ value can effectively balance the regularization intensity, preventing both overfitting and underfitting.

Furthermore, an analysis of the curve distribution characteristics reveals that the model exhibits low sensitivity to γ . Within the optimal range of $[0.001, 0.01]$, the performance fluctuates by less than 1%, thereby verifying the robustness of the TF-GNN model.

5. Conclusion

This paper introduces TF-GNN, a time-aware dynamic group attention network tailored for fraud detection on heterogeneous dynamic graphs. By integrating the Temporal Feedback Manager (TFM) with multi-scale LSTMs, dynamic grouping aggregation, multi-channel attention fusion, and discriminative regularization, TF-GNN effectively addresses limitations in capturing temporal fraud evolution, relational semantics, and noise in existing methods. The two-stage training strategy ensures stable convergence through decoupled temporal and structural optimization. Experiments on four real-world datasets demonstrate TF-GNN's superiority, outperforming baselines by 0.59%–2.25% in AUC and AP, with enhanced robustness under perturbations. Qualitative analyses, including t-SNE visualizations and temporal trajectories, validate its interpretability and fraud pattern discernment. Despite these advances, TF-GNN exhibits higher computational overhead and dependence on historical sequence length. Future directions include extensions to unsupervised paradigms, federated learning, and real-time online detection to enhance practical deployment in financial security systems. Overall, TF-GNN provides a robust, interpretable framework with substantial potential for dynamic fraud detection applications.

CRedit authorship contribution statement

WanPeng Zhen: Writing – original draft, Methodology. **Qi Dai:** Writing – review & editing. **Lifang Chen:** Writing – review & editing. **DongMei Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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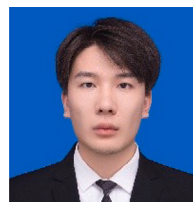
Data availability

Data will be made available upon request.

References

- [1] Z. Liu, Y. Dou, P.S. Yu, Y. Deng, H. Peng, Alleviating the inconsistency problem of applying graph neural network to fraud detection, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2020, pp. 1569–1572.
- [2] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, M. Wang, LightGCN: simplifying and powering graph convolution network for recommendation, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2020, pp. 639–648.
- [3] R. Ying, R. He, K. Chen, P. Eksombatchai, W.L. Hamilton, J. Leskovec, Graph convolutional neural networks for web-scale recommender systems, in: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2018, pp. 974–983.
- [4] D. Marcheggiani, I. Titov, Encoding sentences with graph convolutional networks for semantic role labeling, in: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, 2017, pp. 1506–1515, <https://doi.org/10.18653/v1/D17-1159>
- [5] L. Yao, C. Mao, Y. Luo, Graph convolutional networks for text classification, in: Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence (AAAI-19), AAAI Press, 2019, pp. 7370–7377, <https://doi.org/10.1609/aaai.v33i01.33017370>
- [6] T.N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: ICLR (Poster), 2017, pp. 2713–2726, <https://openreview.net/forum?id=SJU4ayYgl>
- [7] W. Hamilton, Z. Ying, J. Leskovec, Inductive representation learning on large graphs, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [8] P. Velicković, G. Cucurull, A. Casanova, A. Romero, P. Liò, Y. Bengio, Graph attention networks, in: International Conference on Learning Representations (ICLR), 2018, pp. 2920–2931.
- [9] Z. Chai, S. You, Y. Yang, S. Pu, J. Xu, H. Cai, W. Jiang, Can abnormality be detected by graph neural networks? in: Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI), 2022, pp. 1945–1951.
- [10] J. Tang, J. Li, Z. Gao, J. Li, Rethinking graph neural networks for anomaly detection, in: International Conference on Machine Learning, PMLR, 2022, pp. 21076–21089.
- [11] Y. Gao, X. Wang, X. He, Z. Liu, H. Feng, Y. Zhang, Addressing heterophily in graph anomaly detection: a perspective of graph spectrum, in: Proceedings of the ACM Web Conference 2023, 2023, pp. 1528–1538.
- [12] Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, J. Jia, Y. Xiao, J. Wu, H. Guo, J. Han, Enhancing graph neural network-based fraud detectors against camouflaged fraudsters, in: Proceedings of the 29th ACM International Conference on Information and Knowledge Management, CIKM '20, Association for Computing Machinery, New York, NY, USA, 2020, pp. 315–324, <https://doi.org/10.1145/3340531.3411894>.
- [13] Y. Liu, X. Ao, Z. Qin, J. Chi, J. Feng, H. Yang, Q. He, Pick and choose: a GNN-based imbalanced learning approach for fraud detection, in: Proceedings of the Web Conference 2021, 2021, pp. 3168–3177, <https://doi.org/10.1145/3442381.3449989>
- [14] H. Peng, R. Zhang, Y. Dou, R. Yang, J. Zhang, P.S. Yu, Reinforced neighborhood selection guided multi-relational graph neural networks, *ACM Trans. Inf. Syst. (TOIS)* 40 (4) (2021) 1–46, <https://doi.org/10.1145/3490181>
- [15] F. Shi, Y. Cao, Y. Shang, Y. Zhou, C. Zhou, J. Wu, H2-FDetector: a GNN-based fraud detector with homophilic and heterophilic connections, in: Proceedings of the ACM Web Conference 2022, 2022, pp. 1486–1494.
- [16] M. Duan, T. Zheng, Y. Gao, G. Wang, Z. Feng, X. Wang, DGA-GNN: dynamic grouping aggregation GNN for fraud detection, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, 2024, pp. 11820–11828.
- [17] L. Akoglu, H. Tong, D. Koutra, Graph based anomaly detection and description: a survey, *Data Min. Knowl. Discov.* 29 (3) (2015) 626–688.
- [18] X. Ma, J. Wu, S. Xue, J. Yang, C. Zhou, Q.Z. Sheng, H. Xiong, L. Akoglu, A comprehensive survey on graph anomaly detection with deep learning, *IEEE Trans. Knowl. Data Eng.* 35 (12) (2021) 12012–12038.
- [19] Y. Wang, J. Zhang, Z. Huang, W. Li, S. Feng, Z. Ma, Y. Sun, D. Yu, F. Dong, J. Jin, B. Wang, J. Luo, Label information enhanced fraud detection against low homophily in graphs, in: Proceedings of the ACM Web Conference 2023, 2023, <https://api.semanticscholar.org/CorpusID:257050274>.
- [20] J. Tang, H. Gu, D.B. Vuković, G. Xu, Y. Wang, H. Tao, J. Cao, Fraud detection in multi-relation graph: contrastive learning on feature and structural levels, *Neurocomputing* 637 (2025) 130063, <https://api.semanticscholar.org/CorpusID:277334539>.
- [21] J. Qian, G. Tong, Metapath-guided graph neural networks for financial fraud detection, *Comput. Electr. Eng.* 126 (2025) 110428, <https://api.semanticscholar.org/CorpusID:278983286>.
- [22] A. Deng, B. Hooi, Graph neural network-based anomaly detection in multivariate time series, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, 2021, pp. 4027–4035, <https://doi.org/10.1609/aaai.v35i5.16523>
- [23] S. Motie, B. Raahemi, Financial fraud detection using graph neural networks: a systematic review, *Expert Syst. Appl.* 240 (2023) 122156, <https://api.semanticscholar.org/CorpusID:264818923>.
- [24] S. Sharmin, Z. Zaman, SPAM detection in social media employing machine learning tool for text mining, in: 2017 13th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS), 2017, pp. 137–142, <https://api.semanticscholar.org/CorpusID:4873911>.
- [25] R. Mohawesh, H.B. Salameh, Y. Jararweh, M.S. Alkhalaileh, S. Maqsood, Fake review detection using transformer-based enhanced LSTM and Roberta, *Int. J. Cogn. Comput. Eng.* 5 (2024) 250–258.
- [26] N. Chen, Z. Liu, B. Hooi, B. He, R. Fathony, J. Hu, J. Chen, Consistency training with learnable data augmentation for graph anomaly detection with limited supervision, in: International Conference on Learning Representations, 2024, pp. 57506–57533.
- [27] F. Xu, N. Wang, H. Wu, X. Wen, X. Zhao, H. Wan, Revisiting graph-based fraud detection in sight of heterophily and spectrum, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, 2024, pp. 9214–9222, <https://doi.org/10.1609/aaai.v38i8.28773>
- [28] W. Zhuo, Z. Liu, B. Hooi, B. He, G. Tan, R. Fathony, J. Chen, Partitioning message passing for graph fraud detection, in: The Twelfth International Conference on Learning Representations, 2024, pp. 56840–56858, <https://openreview.net/forum?id=tEgrUrUuWA>.
- [29] A. Pareja, G. Domeniconi, J. Chen, T. Ma, T. Suzumura, H. Kanezashi, T. Kaler, C.E. Leiserson, EvolveGCN: evolving graph convolutional networks for dynamic graphs, *ArXiv abs/1902.10191*, 2019, <https://api.semanticscholar.org/CorpusID:67856459>.
- [30] E. Rossi, B.P. Chamberlain, F. Frasca, D. Eynard, F. Monti, M.M. Bronstein, Temporal graph networks for deep learning on dynamic graphs, *ArXiv abs/2006.10637*, 2020, <https://api.semanticscholar.org/CorpusID:219792342>.
- [31] Y. Xie, G. Liu, M. Zhou, L. Wei, H. Zhu, R. Zhou, L. Cao, A spatial-temporal gated network for credit card fraud detection by learning transactional representations, *IEEE Trans. Autom. Sci. Eng.* 21 (4) (2023) 6978–6991.
- [32] Y. Tian, G. Liu, Spatial-temporal-aware graph transformer for transaction fraud detection, *IEEE Trans. Ind. Inform.* 20 (11) (2024) 12659–12668.
- [33] C. Yang, H. Liu, D. Wang, Z. Zhang, C. Yang, C. Shi, FLAG: fraud detection with LLM-enhanced graph neural network, in: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining v. 2, 2025, pp. 5150–5160.
- [34] Y. Li, J. Hu, B. Hooi, B. He, C. Chen, DGP: a dual-granularity prompting framework for fraud detection with graph-enhanced LLMs, *arXiv preprint*, 2025 [arXiv:2507.21653](https://arxiv.org/abs/2507.21653).
- [35] M. Weber, G. Domeniconi, J. Chen, D.K.I. Weidele, C. Bellei, T. Robinson, C.E. Leiserson, Anti-money laundering in bitcoin: experimenting with graph convolutional networks for financial forensics, *ArXiv abs/1908.02591*, 2019, <https://api.semanticscholar.org/CorpusID:199472611>.
- [36] S. Rayana, L. Akoglu, Collective opinion SPAM detection: bridging review networks and metadata, in: Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2015, <https://api.semanticscholar.org/CorpusID:14551027>.
- [37] J. McAuley, J. Leskovec, From amateurs to connoisseurs: modeling the evolution of user expertise through online reviews, in: Proceedings of the 22nd International Conference on World Wide Web, 2013, <https://api.semanticscholar.org/CorpusID:3047006>.
- [38] G.E. Hinton, Connectionist learning procedures, *Artif. Intell.* 40 (1989) 185–234, <https://api.semanticscholar.org/CorpusID:7840452>.
- [39] L. Breiman, Random forests, *Mach. Learn.* 45 (2001) 5–32, <https://api.semanticscholar.org/CorpusID:89141>.
- [40] W. Zhuo, Z. Liu, B. Hooi, B. He, G. Tan, R. Fathony, J. Chen, Partitioning message passing for graph fraud detection, in: The Twelfth International Conference on Learning Representations (ICLR), 2024, <https://openreview.net/forum?id=tEgrUrUuWA>

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